🛠️ Tech Stack

This project is built using a modern, open-source technology stack designed for efficient audio processing, Retrieval-Augmented Generation (RAG), and AI-powered question answering. Each tool was selected based on reliability, performance, ease of integration, and availability of free or local options.

Tool	Purpose	Why This Tool?
yt-dlp	Downloads audio from YouTube videos	Free, reliable, actively maintained, and supports a wide range of video platforms.
FFmpeg	Extracts audio from video files	Industry-standard multimedia processing tool with excellent performance and flexibility.
OpenAI Whisper	Transcribes English audio into text	Open-source speech-to-text model that provides high-quality transcription and runs locally.
Sarvam AI API	Transcribes Hindi and Hinglish audio	One of the best free APIs for accurate Hindi and Hinglish speech recognition.
LangChain	Connects all AI pipeline components	Simplifies the creation of Retrieval-Augmented Generation (RAG) workflows and tool orchestration.
Hugging Face Embeddings	Converts text into vector embeddings	Free, open-source, supports local execution, and produces high-quality semantic embeddings.
ChromaDB	Stores and retrieves vector embeddings	Lightweight, free, easy to set up, and optimised for semantic search applications.
rank-bm25	Performs keyword-based document retrieval	Fast, lightweight, and improves retrieval by matching exact keywords.
CrossEncoder	Reranks retrieved documents	Enhances retrieval precision by selecting the most relevant context before answer generation.
Mistral AI	Generates responses from retrieved context	Offers a powerful language model with a free API suitable for RAG applications.
RAGAS	Evaluates the RAG system	Widely used research framework for measuring answer quality, retrieval accuracy, and faithfulness.
Streamlit	Builds the web user interface	Enables rapid development of clean, interactive web applications with minimal code.

Why This Tech Stack?

- ✅ Completely free or offers generous free-tier usage for development.
- ✅ Supports both English and Hindi/Hinglish transcription.

- ✅ Uses Retrieval-Augmented Generation (RAG) for more accurate responses.
- ✅ Modular architecture makes it easy to upgrade or replace individual components.
- ✅ Suitable for research projects, academic work, and production-ready AI assistants.
- ✅ Fast to develop, easy to maintain, and scalable for future enhancements.